

DBMS - 7

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mysql> use ish;

Reading table information for completion of table and column names

You can turn off this feature to get a quicker startup with -A

Database changed

mysql> create table Library(Book_Id int(5) not null primary key, Book_name varchar(50), Author varchar(50), Price int(5), Quantity int(5));

Query OK, 0 rows affected, 3 warnings (0.02 sec)

mysql> insert into Library(Book_Id, Book_name, Author, Price, Quantity);

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near " at line 1

mysql> insert into Library(Book_Id, Book_name, Author, Price, Quantity)

-> values(101, "The Power Of your subconscious mind", "Joseph Murphy", 149 , 2),

-> (102, "Rich Dad Poor Dad", "Robert Kiyosaki" , 500, 1),

-> (103, "One night @ the call center", "Chetan Bhagat", 525, 1),

-> (104, "Journey To Atlantis", "Thea Stilton", 700, 1),

-> (105, "Who Stole My Cheese?!", "Geronimo Stilton", 825, 7);

Query OK, 5 rows affected (0.00 sec)

Records: 5 Duplicates: 0 Warnings: 0

mysql> create table Library_Audit(Lib_Id int(5) not null primary key auto_increment , Book_Id int(5), Book_name varchar(50), Author varchar(50), Price int(5), Quantity int(5), Operation_Type varchar(50),

Operation_Time datetime);

Query OK, 0 rows affected, 4 warnings (0.02 sec)

mysql> DELIMITER //

mysql> CREATE TRIGGER before_library_update

-> BEFORE UPDATE ON Library

-> FOR EACH ROW

-> BEGIN

-> INSERT INTO Library_Audit

-> VALUES (

-> OLD.Book_Id,

-> OLD.Book_name,

-> OLD.Author,

```

-> OLD.Price,
-> OLD.Quantity,
-> 'before_update',
-> NOW()
-> );
-> END //

```

Query OK, 0 rows affected (0.01 sec)

```

mysql>
mysql> DELIMITER ;

```

```

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER after_library_update
-> AFTER UPDATE ON Library
-> FOR EACH ROW
-> BEGIN
-> INSERT INTO Library_Audit (Book_Id, Book_name, Author, Price, Quantity,
Operation_Type, Operation_Time)
-> VALUES (
-> NEW.Book_Id, -- Changed from OLD to NEW
-> NEW.Book_name, -- Changed from OLD to NEW
-> NEW.Author, -- Changed from OLD to NEW
-> NEW.Price, -- This will now show 600
-> NEW.Quantity, -- Changed from OLD to NEW
-> 'after_update',
-> NOW()
-> );
-> END //

```

Query OK, 0 rows affected (0.00 sec)

```

mysql>
mysql> DELIMITER ;
mysql> UPDATE Library SET Price = 650 WHERE Book_Id = 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

mysql> SELECT * FROM Library_Audit;

```

```

+-----+-----+-----+-----+-----+-----+
+-----+
| Lib_Id | Book_Id | Book_name | Author | Price | Quantity | Operation_Type |
| Operation_Time |

```

```

+-----+-----+-----+-----+-----+-----+-----+
-----+
| 1 | 101 | The Power Of your subconscious mind | Joseph Murphy | 149 | 2 |
before_update | 2026-02-27 11:38:19 |
| 2 | 101 | The Power Of your subconscious mind | Joseph Murphy | 149 | 2 |
after_update | 2026-02-27 11:38:19 |
| 3 | 101 | The Power Of your subconscious mind | Joseph Murphy | 600 | 2 |
before_update | 2026-02-27 11:40:42 |
| 4 | 101 | The Power Of your subconscious mind | Joseph Murphy | 650 | 2 |
after_update | 2026-02-27 11:40:42 |
+-----+-----+-----+-----+-----+-----+-----+
-----+
4 rows in set (0.00 sec)

```

```

mysql> CREATE TRIGGER after_library_update
-> AFTER UPDATE ON Library
-> FOR EACH ROW
-> BEGIN
-> INSERT INTO Library_Audit (Book_Id, Book_name, Author, Price, Quantity,
Operation_Type, Operation_Time)
-> VALUES
-> (
-> NEW.Book_Id,
-> NEW.Book_name,
-> NEW.Author,
-> NEW.Price,
-> NEW.Quantity,
-> 'after_update',
-> NOW()
-> );
-> END //

```

Query OK, 0 rows affected (0.01 sec)

```

mysql> DELIMITER ;
mysql> UPDATE Library SET Price = 250 WHERE Book_Id = 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

mysql> SELECT * FROM Library_Audit;

```

```

+-----+-----+-----+-----+-----+-----+-----+
-----+

```

Lib_Id	Book_Id	Book_name	Author	Price	Quantity	Operation_Type	Operation_Time
1	101	The Power Of your subconscious mind	Joseph Murphy	149	2		
before_update 2026-02-27 11:38:19							
2	101	The Power Of your subconscious mind	Joseph Murphy	149	2		
after_update 2026-02-27 11:38:19							
3	101	The Power Of your subconscious mind	Joseph Murphy	600	2		
before_update 2026-02-27 11:40:42							
4	101	The Power Of your subconscious mind	Joseph Murphy	650	2		
after_update 2026-02-27 11:40:42							
5	101	The Power Of your subconscious mind	Joseph Murphy	650	2		
before_update 2026-02-27 11:49:10							
6	101	The Power Of your subconscious mind	Joseph Murphy	250	2		
after_update 2026-02-27 11:49:10							

6 rows in set (0.00 sec)

```

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER before_library_delete
-> BEFORE DELETE ON Library
-> FOR EACH ROW
-> BEGIN
->   INSERT INTO Library_Audit (
->     Book_Id,
->     Book_name,
->     Author,
->     Price,
->     Quantity,
->     Operation_Type,
->     Operation_Time
->   )
->   VALUES (
->     OLD.Book_Id,
->     OLD.Book_name,
->     OLD.Author,
->     OLD.Price,
->     OLD.Quantity,
->     'before_delete',

```

```
-> NOW()
-> );
-> END //
```

Query OK, 0 rows affected (0.00 sec)

mysql>

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE TRIGGER after_library_delete
```

```
-> AFTER DELETE ON Library
-> FOR EACH ROW
-> BEGIN
->   INSERT INTO Library_Audit(
->     Book_Id,
->     Book_name,
->     Author,
->     Price,
->     Quantity,
->     Operation_Type,
->     Operation_Time
->   )
->
->
-> VALUES (
->   OLD.Book_Id,
->   OLD.Book_name,
->   OLD.Author,
->   OLD.Price,
->   OLD.Quantity,
->   'after_delete',
->   NOW()
-> );
-> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql> DELETE FROM Library WHERE Book_Id = 101;
```

Query OK, 1 row affected (0.00 sec)

```
mysql> SELECT * FROM Library_Audit;
```

```

+-----+-----+-----+-----+-----+-----+-----+
-----+
| Lib_Id | Book_Id | Book_name                | Author   | Price | Quantity | Operation_Type
| Operation_Time   |
+-----+-----+-----+-----+-----+-----+-----+
-----+
|   1 |   101 | The Power Of your subconscious mind | Joseph Murphy | 149 | 2 |
before_update | 2026-02-27 11:38:19 |
|   2 |   101 | The Power Of your subconscious mind | Joseph Murphy | 149 | 2 |
after_update  | 2026-02-27 11:38:19 |
|   3 |   101 | The Power Of your subconscious mind | Joseph Murphy | 600 | 2 |
before_update | 2026-02-27 11:40:42 |
|   4 |   101 | The Power Of your subconscious mind | Joseph Murphy | 650 | 2 |
after_update  | 2026-02-27 11:40:42 |
|   5 |   101 | The Power Of your subconscious mind | Joseph Murphy | 650 | 2 |
before_update | 2026-02-27 11:49:10 |
|   6 |   101 | The Power Of your subconscious mind | Joseph Murphy | 250 | 2 |
after_update  | 2026-02-27 11:49:10 |
|   7 |   101 | The Power Of your subconscious mind | Joseph Murphy | 250 | 2 |
before_delete | 2026-02-27 12:04:40 |
|   8 |   101 | The Power Of your subconscious mind | Joseph Murphy | 250 | 2 |
after_delete  | 2026-02-27 12:04:40 |
+-----+-----+-----+-----+-----+-----+-----+
-----+
8 rows in set (0.00 sec)

```
